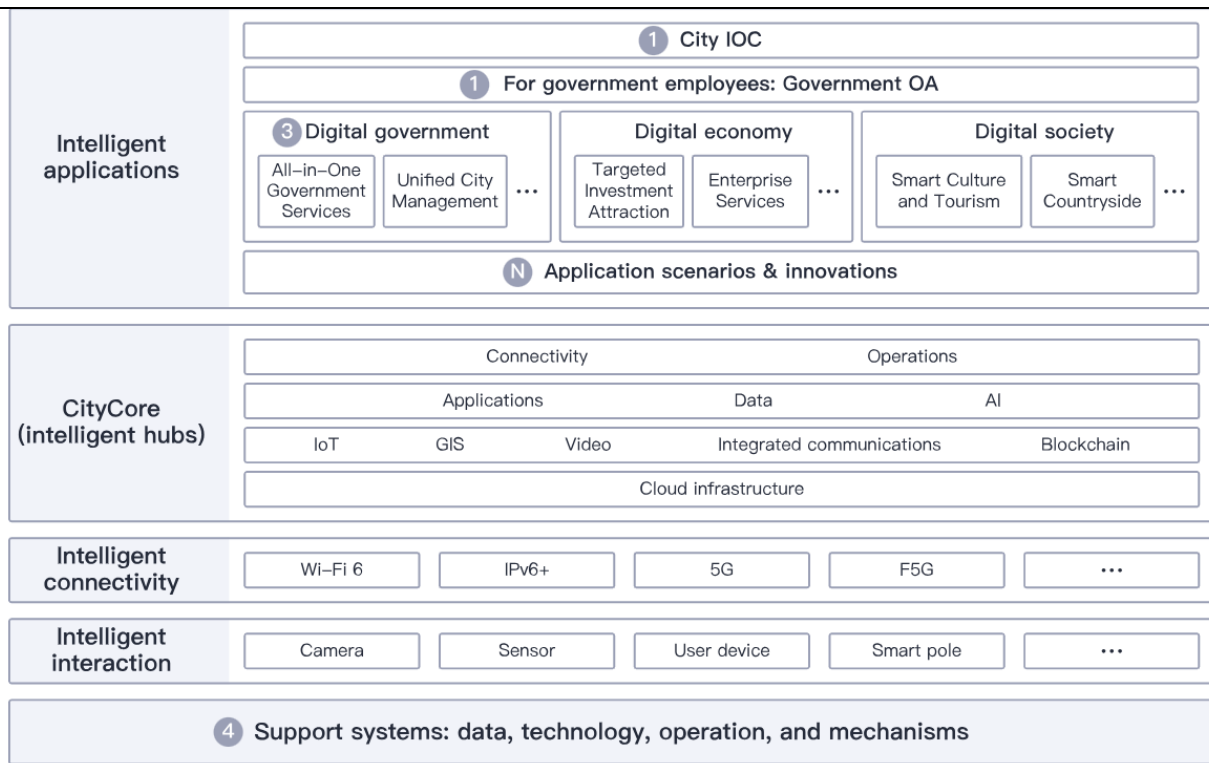


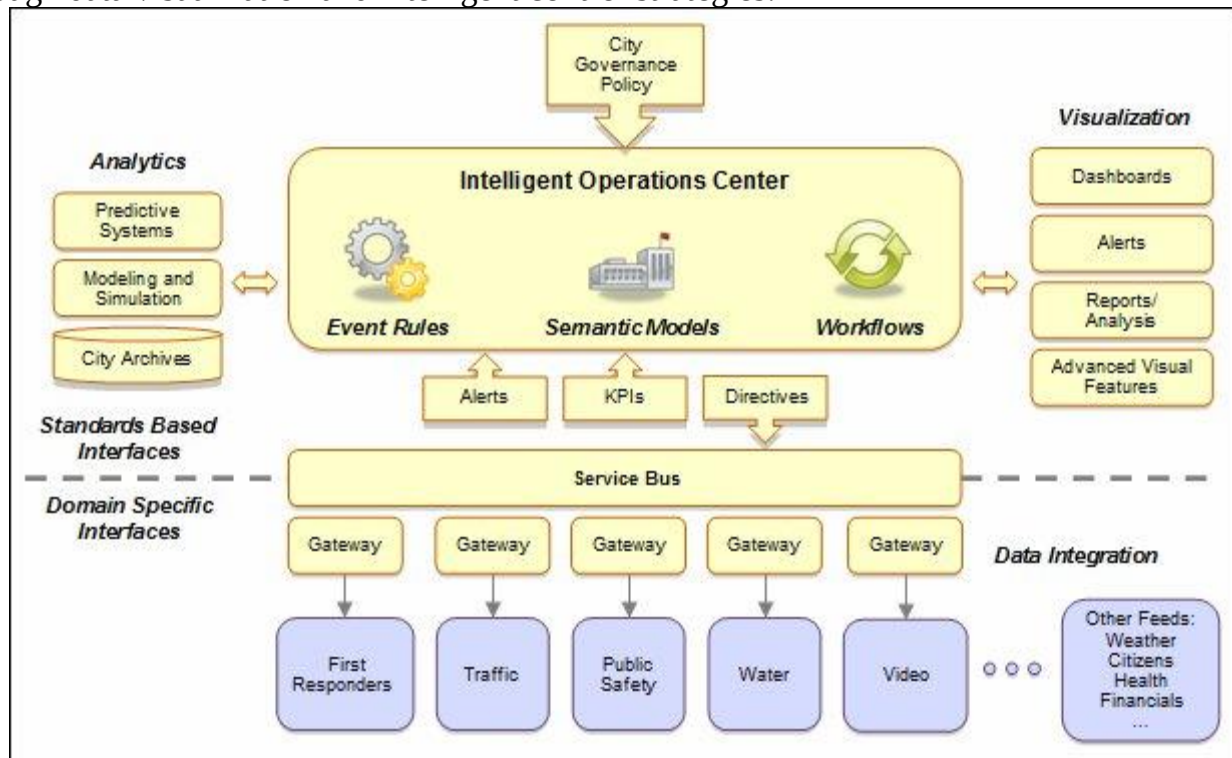
## INTERNSHIP PROPOSAL

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Research Team or Component</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |                                                                                                                             |
| <b>Supervisors</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Name/Nom     | Van Lic TRAN                                                                                                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Title /Titre | Dr.                                                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Telephone    | (+84)706145815                                                                                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | E-mail       | tvlic@dut.udn.vn                                                                                                            |
| <b>Workplace</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |                                                                                                                             |
| <b>Internship Category</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              | IUT/Bachelor <input type="checkbox"/> Master1/Engineer4 <input type="checkbox"/> Master2/Engineer5 <input type="checkbox"/> |
| <b>Number of students and specialties (desired pair)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              | 2                                                                                                                           |
| <b>Student's name and contact information (if known)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                                                                                                                             |
| <b>Proposal title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              | <b>Design and Development of an Intelligent Operation Center (IOC) for Building Energy Management and Automated Control</b> |
| <p style="text-align: center;"><b>Description of the proposal (2-3 pages max)</b></p> <p><b>1. Introduction:</b><br/>Energy consumption in modern buildings continues to increase due to the rising demand for comfort, facility operations, and interconnected digital systems. Inefficient energy usage not only raises operational costs but also affects sustainability objectives and environmental impact. To address these challenges, intelligent building management solutions have become essential, enabling centralized monitoring, real-time control, and data-driven optimization. An Intelligent Operation Center (IOC) provides a unified platform that integrates sensors, actuators, HVAC systems, lighting networks, and power meters into a cohesive management interface. Through advanced data analytics and automated control strategies, the IOC helps reduce energy waste, enhance equipment lifespan, and improve overall building performance. This project focuses on designing and developing an IOC that supports real-time monitoring, anomaly detection, automated decision-making, and efficient energy-management operations.</p> |              |                                                                                                                             |



## 2. Objective:

The primary objective of this internship is to develop a functional Intelligent Operation Center (IOC) capable of monitoring building energy usage and performing automated control of key energy-consumption systems. The project aims to integrate IoT-based sensing devices, data acquisition modules, and control units into a centralized dashboard that provides real-time insights and automation capabilities. The final IOC prototype will help optimize energy efficiency, ensure safe operation of building systems, and assist facility managers in making informed decisions through data visualization and intelligent control strategies.



## 3. Work to be developed in this internship:

The internship will involve several essential development tasks. First, the student will conduct research on building energy-management architectures, communication protocols, and existing IOC solutions. Second, a sensing and control infrastructure will be designed, including the selection

of power meters, environmental sensors, smart switches, and communication interfaces. Third, the student will develop the IOC software platform, consisting of data collection modules, a real-time monitoring dashboard, and an automated control engine. Key functionalities include energy visualization, alert notifications, device scheduling, and rule-based or AI-assisted control. Fourth, system integration and testing will be performed to evaluate performance metrics such as response time, data accuracy, reliability, and energy-saving effectiveness. Finally, a demonstration scenario will be prepared to showcase the IOC's capabilities in a simulated building environment.

**4. Specific technic tools and working conditions:**

The project will utilize IoT hardware such as smart energy meters, environmental sensors (temperature, humidity, and occupancy), and actuator modules (relays, smart switches, and HVAC controllers). Communication technologies may include Modbus, BACnet, MQTT, or LoRaWAN depending on system requirements. The IOC software platform will be developed using tools such as Node-RED, Python, Grafana, or a web-based UI framework like React. A database system such as PostgreSQL, InfluxDB, or Firebase will be used for storing and analyzing energy data. Development work will be conducted in a laboratory environment equipped with hardware testing stations, local server infrastructure, measurement tools, and controlled electrical loads for system validation.

**5. Documentation references:**

Documentation for this project includes technical literature on Building Energy Management Systems (BEMS), smart building automation, and IoT-based control platforms. Additional references include communication protocol standards (Modbus, BACnet, MQTT), sensor and actuator datasheets, energy-efficiency guidelines, and best practices in intelligent facility management. Relevant textbooks, manufacturer manuals, research papers on energy optimization, and system-integration guides will also serve as essential resources for the development process.

**6. Appendix (if necessary):**